

Interface Dimensionality and the Independence of Generative Capacity:
Why Some Constraints Are Fertile and Others Are Sterile

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Abstract

This paper introduces *interface dimensionality* (I_6) as the candidate mechanism underlying the proposed variable Γ (Generative Capacity) within the Theory of Non-Equilibrium Dissipative Systems. Interface dimensionality is defined as the number of compositionally distinct contexts in which a structure can participate as a component, where distinctness is measured by the constraint rules of the architecture rather than by observed participation history. The paper argues that I_6 resolves three outstanding problems in the Γ lineage: the circularity of the reusability criterion, the representation dependence of the admissible structure count N_a , and the unresolved relationship between Γ and the constraint satisfaction reserve S^c from the TNDS-XR framework. The central theoretical claim is that Γ and S^c are genuinely independent variables measuring orthogonal architectural properties: S^c governs the depth of a structure's stability well, while I_6 governs the breadth of its compositional horizon. The paper further proposes a surface-constraint principle: constraints generate high I_6 when they operate at the interface layer of a structure rather than at its internal composition layer, simultaneously enforcing reproductive specificity and preserving contextual abstractness. This principle is demonstrated across chemistry, language, mathematics, and technology, and generates a specific falsifiable prediction: among systems with matched complexity, entropy, modularity, and S^c , the system with higher average I_6 will generate more compositional layers per unit of evolutionary or developmental time.

Keywords: interface dimensionality, generative capacity, surface constraint, compositional fertility, TNDS, emergence, evolutionary evolvability, constraint architecture

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Introduction

The observation that some physical and formal architectures generate successive layers of increasing structural complexity while others remain compositionally inert is one of the oldest unresolved problems in the study of emergence. Chemistry generates biology. Biology generates cognition. Cognition generates language. Language generates mathematics and scientific institutions. Crystal lattices, planetary systems, and stable rocks do not participate in these transitions despite being thermodynamically robust, physically complex, and energetically maintained. The question of what distinguishes the fertile architectures from the sterile ones has resisted unified treatment because the standard explanatory variables, complexity, entropy, evolvability, modularity, and stability, each capture one aspect of the phenomenon while leaving the others unaddressed.

The Theory of Non-Equilibrium Dissipative Systems (TNDS), developed across a series of prior audits, has approached this problem through successive compression of candidate primitives. The most recent stage of this development proposed the variable Γ , Generative Capacity, defined as the fractional growth rate of admissible composite structures per unit of compositional depth. A hostile adjudication of Γ identified three unresolved problems: circularity in the definition of a reusable structure, representation dependence of the admissible structure count, and an unresolved relationship between Γ and the constraint satisfaction reserve S° from the TNDS-XR framework.

This paper addresses all three problems by proposing a candidate mechanism for Γ and arguing that this mechanism, interface dimensionality, is both conceptually prior to Γ and sufficient to resolve the outstanding issues. The paper is organized as follows. Section 2 defines interface dimensionality precisely and demonstrates its non-circularity. Section 3 argues that Γ

and S° are independent variables measuring orthogonal architectural properties. Section 4 develops the surface-constraint principle, which explains why some constraints generate high interface dimensionality while others do not. Section 5 applies the principle across four domains. Section 6 derives falsifiable predictions and proposes a research program. Section 7 addresses the most serious remaining objections.

Interface Dimensionality: Definition and Non-Circularity

The central problem with the original Γ formulation was that reusability was defined by reference to future participation in the architecture's generative history. A structure was counted as reusable if it was, in fact, reused. This made Γ a retrospective descriptor of emergence rather than a predictor of it. The variable measured what had already happened rather than predicting what would happen next.

Interface dimensionality resolves this circularity by grounding the definition in present architectural structure rather than future generative history. The formal definition is as follows.

Definition 1 (Interface Dimensionality). Let S be a structure generated by an architecture A . The interface dimensionality $I_6(S, A)$ is the number of compositionally distinct contexts in which S can participate as a component, where two contexts are compositionally distinct if and only if the resulting composite structures have different properties under the constraint rules of A , and where compositional admissibility is determined by the constraint rules of A applied to S 's current form, without reference to S 's generative history or future participation.

Three features of this definition are worth making explicit. First, admissibility is determined by constraint rules applied to current form. This means that whether a structure can combine with another is determined by the physical, formal, or syntactic properties of the two structures as they currently exist, not by whether they have combined before or will combine in the future. The admissible bonds available to a carbon atom are determined by its valence configuration, not by its history. This breaks the circularity.

Second, the distinctness criterion is property-based rather than identity-based. Two compositional contexts are distinct if they produce compositionally different results, not merely if they involve structurally different partners. This matters because a structure with high interface dimensionality can combine with many different partners to produce many functionally different composite structures, whereas a structure with low interface dimensionality produces compositionally equivalent results regardless of which partner it combines with.

Third, the definition is local to an architecture. Interface dimensionality is always measured relative to a specific constraint system. The same physical object may have different interface dimensionality values relative to different constraint vocabularies. A protein's interface dimensionality relative to biochemical constraint rules differs from its interface dimensionality relative to gravitational constraint rules. This is a feature rather than a flaw because it correctly identifies that structural fertility is architecture-relative.

With this definition in place, Γ can be reformulated as:

$$\Gamma = \frac{d\mathbb{I}_6(L)}{dL}$$

where $\mathbb{I}_6(L)$ is the average interface dimensionality of structures generated at compositional depth L . This formulation is non-circular because $\mathbb{I}_6$ is defined on current structural properties, not on generative outcomes. It is also more directly interpretable: Γ now

measures how rapidly the average compositional horizon of newly generated structures increases as compositional depth increases.

The Independence of Γ from S^c

The most important theoretical question in the Γ lineage is whether generative capacity is an independent variable or a consequence of the constraint satisfaction reserve already formalized in the TNDS-XR architecture. The TNDS-XR framework established the master dynamics:

$$dS^c/dt = (W_k\eta/L) - (G_e + G_i)$$

where S^c is the constraint satisfaction reserve, W_k is corrective work capacity, η is correction efficiency, L is latency, G_e is external divergence, and G_i is internal divergence. This equation governs whether a structure persists. The question is whether Γ is determined by S^c or independent of it.

The argument for dependence runs as follows. Systems with high S^c have surplus resources available for exploration. Exploratory behavior generates new structural configurations. New structural configurations are what Γ measures. Therefore Γ is a downstream consequence of S^c and not an independent variable.

This argument fails for a specific reason that the interface dimensionality formulation makes visible. S^c measures the depth of a structure's stability well: how far the structure can be perturbed before losing its identity. I_6 measures the breadth of a structure's compositional horizon: how many distinct contexts the structure can enter as a component while maintaining that identity. These are genuinely orthogonal dimensions of architectural character.

Four quadrant cases demonstrate the independence:

	Low I_6	High I_6	
High S°	Crystal lattice. Deep stability well, narrow compositional horizon. Persistent but sterile.	Folded protein domain. Deep stability, broad binding surface. Persistent and fertile.	
Low S°	Random polymer. Shallow stability, narrow composition. Transient and sterile.	RNA aptamer under prebiotic stress. Shallow stability, broad template function. Transient but fertile.	

The existence of all four quadrant cases is the decisive argument for independence. A crystal lattice has enormous S° and near-zero I_6 . A prebiotic RNA aptamer may have low S° and high I_6 . If Γ were a consequence of S° , we would expect high S° to predict high I_6 . The crystal case falsifies this prediction directly. Thermodynamic stability is neither necessary nor sufficient for compositional fertility.

The independence can also be shown analytically. S° is determined by the energy landscape of the structure relative to its environment: how deep its basin of attraction is relative to the thermal noise of the system. I_6 is determined by the geometric and chemical properties of the structure's surface: how many distinct binding modes it presents to other structures. These are different physical quantities with different measurement procedures and different natural scales. Their correlation is empirically contingent rather than structurally necessary.

This establishes that the TNDS-XR architecture requires an extension. The existing framework accounts for the persistence of structures under divergence. It does not account for their compositional fertility once they persist. The two variables together, S° governing persistence and I_6 governing fertility, provide a more complete architectural description than either alone.

The Surface-Constraint Principle

The identification of interface dimensionality as the mechanism underlying Γ immediately generates a new question: what determines I_6 ? Why do some constraints generate structures with high interface dimensionality while others do not?

The answer proposed here is the surface-constraint principle. The principle states that constraints generate high interface dimensionality when they operate at the interface layer of a structure rather than at its internal composition layer. More precisely, a constraint is surface-operative if it governs how a structure combines with other structures, without specifying what those structures must be. A constraint is deep-operative if it governs the internal composition of a structure, specifying what it is made of rather than how it interacts.

The Surface-Constraint Principle. A constraint C operating on an architecture A generates high average interface dimensionality in the structures it produces when C is surface-operative: when C restricts the modes of combination available at the output's boundary without restricting the identity of the entities that can occupy those modes. Conversely, a constraint that is deep-operative, restricting internal composition without creating new surface interaction modes, generates low or neutral average interface dimensionality.

The principle explains the difference between the crystal and the protein. Crystallographic symmetry constraints are deep-operative: they determine the internal arrangement of atoms at lattice positions. The resulting surface presents a geometrically regular pattern with a small vocabulary of interaction modes. The constraint is precise but internally directed. Protein folding constraints, by contrast, include surface-operative components: the requirements for hydrophobic core burial create an aqueous-facing surface whose specific topology is not fully determined by the folding constraints themselves. The constraint specifies

what must be inside but leaves the surface topology highly variable. That variability is what generates diverse binding interfaces.

The principle also explains why adding layers of constraint can increase rather than decrease interface dimensionality. When a new constraint operates at the surface of structures generated by a prior constraint layer, it creates new categories of interaction modes that did not exist at the prior layer. The new constraint does not merely eliminate possibilities; it reorganizes the interaction vocabulary of the system's structures.

Four-Domain Application

Chemistry

Valence rules are the paradigm case of surface-operative constraints. A valence rule specifies how many bonds an atom can form and of what type, without specifying what atoms must occupy those bond positions. The carbon atom's tetrahedral bonding geometry is a surface constraint: it determines the number and geometric arrangement of interaction modes available, but leaves the identity of binding partners entirely open. This is why carbon-based chemistry generates enormous structural diversity. Every constraint on bond formation is a constraint on how structures can be combined, not on what they must be combined with. The result is a combinatorial explosion in which the generative fertility of each additional compositional layer exceeds the layer before it.

Contrast this with ionic lattice formation. The electrostatic constraint governing ionic crystal formation is deep-operative in the relevant sense: it specifies that opposite charges must alternate at precise geometric positions. The resulting structures have limited surface interaction modes because the surface is simply a continuation of the bulk interior. The constraint is

powerful and produces extraordinarily stable structures, but those structures have small compositional horizons because their surfaces do not present diverse binding modes.

Language

Syntactic constraints provide the clearest non-chemical illustration of the surface-constraint principle. Syntactic rules specify how phrases of different categories can be combined, without specifying what semantic content those phrases must carry. A rule like the Merge operation in minimalist syntax specifies that two syntactic objects can be combined to form a new syntactic object, with the external merge creating a combinatorial space defined by category compatibility rather than by content restriction. The constraint operates at the surface of the combining elements, their syntactic category labels, rather than at their internal composition, their semantic content.

This is why natural language has unbounded productivity. Each grammatical sentence is a new interface-dimensionality-bearing structure: it can be embedded in indefinitely many discourse contexts, quoted, negated, questioned, relativized, and combined with other sentences in content-independent ways. The syntactic constraints that produced the sentence did not constrain what the sentence means, only how it is structured. The meaning determines the sentence's semantic interfaces, which multiply the compositional opportunities at the discourse level. The combination of syntactic surface constraints and open semantic content is what generates the characteristic recursive productivity of natural language.

Mathematics

Mathematical axioms exemplify surface-operative constraints at their purest. An axiom specifies what operations are valid and what properties must be preserved under those

operations, without specifying what mathematical objects must serve as inputs. The group axioms, for instance, specify that a binary operation must be associative, have an identity element, and have inverses, without specifying what the elements are. This surface specificity without content specificity is precisely what makes group theory applicable across seemingly unrelated mathematical domains: number systems, geometric transformations, permutation structures, and topological spaces all satisfy the group axioms despite being internally constructed from very different mathematical materials. The axioms constrain the interface vocabulary of groups without constraining what can appear behind that interface.

The productivity of mathematical formalism is a direct consequence of this principle. A theorem proven within a formal system is a structure with high interface dimensionality because it can participate in indefinitely many subsequent proofs regardless of its subject matter. The proof establishes that certain inferential transitions are valid, and the validity is interface-level: it depends on the logical structure of the argument, not on the specific content of the propositions. This is why mathematical results transfer so effectively across domains.

Scientific Institutions

Scientific institutions provide the most complex application of the principle. An institution like peer review or experimental replication is a surface-operative constraint on knowledge claims: it specifies the conditions under which a claim can be accepted into the scientific record, without specifying what claims must be made. The constraint operates at the boundary between proposed knowledge and accepted knowledge, governing the mode of entry rather than the content of what enters. This is why scientific methodology generates cumulative knowledge growth rather than merely constraining it: the methodological constraints create a

shared interface vocabulary for knowledge claims, making it possible for results from radically different domains to be combined, compared, and built upon.

The surface-constraint principle explains why scientific institutions have higher Γ than individual scientific disciplines. A discipline's constraints are partially content-specific: they specify what questions are legitimate, what methods are appropriate, and what kinds of explanations are acceptable. These are partially deep-operative constraints. Methodological constraints, by contrast, are surface-operative: they govern how any knowledge claim, regardless of content, can be validated and integrated. The methodological layer generates the cross-domain compositional fertility that has historically produced the most significant scientific advances.

Falsifiable Predictions and Research Program

The interface dimensionality proposal generates the following specific falsifiable predictions, ordered by testability.

Prediction 1: Interface Dimensionality Outperforms Complexity in Predicting Architectural Fertility

Among systems matched on structural complexity, Shannon entropy, and S° , the system with higher average interface dimensionality will generate more distinct compositional layers over equivalent time spans. This prediction is testable by constructing paired comparisons in chemistry, linguistics, and mathematical structure. The critical test is the existence of the high- S° / low- I_6 case (crystal-like systems) failing to generate higher-order architecture despite thermodynamic robustness, alongside the low- S° / high- I_6 case (RNA-like systems) generating higher-order architecture despite thermodynamic fragility.

Prediction 2: Surface-Operative Constraints Generate More I_6 per Constraint Layer Than Deep-Operative Constraints

In developmental biology, regulatory networks that operate at the gene expression boundary (surface-operative) should generate more diverse cell types per regulatory module than those that operate on internal metabolic pathways (deep-operative). This prediction is testable using comparative genomics across organisms of varying body plan complexity. The modularity index of transcription factor networks should correlate with body plan I_6 as measured by cell type diversity relative to genome size.

Prediction 3: Representation Invariance of Interface Dimensionality

The interface dimensionality of a protein structure should remain approximately stable across different representational encodings, amino acid sequence, folded geometry, and interaction graph, while the absolute value of N_a (admissible structure count) varies substantially across those same encodings. This prediction is testable by measuring I_6 and N_a for a panel of proteins using three independent representation schemes and comparing the coefficient of variation across representations for the two quantities. A statistically significant difference in representation sensitivity between I_6 and N_a would support the claim that I_6 is the more invariant target variable.

Prediction 4: Phase Transitions in Γ Correspond to Architectural Shifts in I_6

The RNA-to-cellular-life transition should exhibit a non-continuous increase in average I_6 at the point where ribosomal translation machinery emerges. This can be tested by constructing a measure of I_6 for molecular structures at successive stages of prebiotic chemical evolution using known chemical kinetics and comparing the growth rate of I_6 across the

pre-translation and post-translation epochs. A step change rather than a continuous increase in average I_6 at the translation threshold would support the phase-transition hypothesis.

Prediction 5: I_6 Predicts Technological Branching

In technology genealogy networks, the number of descendant applications generated by a technological platform over a 50-year window should correlate more strongly with the I_6 of the platform's core enabling constraint than with the platform's complexity, market size at introduction, or resource investment. The transistor, whose enabling constraint is surface-operative in the sense that junction geometry governs switching behavior independently of circuit content, should outrank the steam engine, whose enabling constraint is partially deep-operative in that cylinder geometry and thermodynamic cycle are inseparable from the specific physical implementation.

Research Program

The most efficient path to testing these predictions combines three research streams. The first is computational: developing formal operationalizations of I_6 in at least two domains, chemistry and linguistics, and measuring I_6 versus S^c across matched structural panels. The second is historical: applying the surface-constraint principle to technology genealogy networks and developmental biology datasets already available in the literature. The third is theoretical: deriving the relationship between the surface-constraint principle and known results in modularity theory, developmental robustness, and linguistic productivity to identify whether I_6 subsumes or extends those frameworks.

The decisive early test is Prediction 3. If interface dimensionality proves to be more representation-invariant than raw structure counts, the measurement program becomes tractable

in a way it was not for the original Γ formulation. This test requires no new data, only the application of three existing protein representation schemes to a standard benchmark panel of well-characterized proteins, and comparison of the resulting I_6 and N_a values.

Objections and Responses

Objection 1: Interface Dimensionality Still Depends on a Constraint System That Must Be Specified

The definition of I_6 is relative to the constraint rules of an architecture A . A critic may argue that specifying A is as difficult as specifying what counts as a reusable structure in the original formulation. This objection has partial force. In well-characterized physical and formal domains, chemistry, syntax, mathematical logic, A is independently specified by the domain's known constraint rules. The challenge of specifying A is the ordinary challenge of domain-specific science, not a special problem for the I_6 proposal. In undercharacterized domains, A must be estimated from structural analysis, which introduces measurement uncertainty but not principled unresolvability. The objection identifies a practical challenge rather than a theoretical defect.

Objection 2: The Surface vs. Deep Distinction Is Unclear in Complex Systems

Real constraint systems typically involve multiple levels simultaneously. Protein folding constraints, for instance, include both surface-operative components (hydrophobic burial creating the aqueous-facing interface) and deep-operative components (secondary structure formation determining the internal configuration). The surface-constraint principle applies to the surface-operative components specifically, not to the constraint system as a whole. A more

precise formulation is that I_6 increases with the proportion of the active constraint set that is surface-operative relative to deep-operative. Complex systems with mixed constraint architectures will show intermediate I_6 values, not binary outcomes. This is empirically tractable and does not undermine the principle.

Objection 3: High I_6 May Be Sufficient but Not Necessary for Generative Fertility

It is conceivable that a system with low I_6 generates higher-order architectural layers through mechanisms not captured by the interface dimensionality framework. The crystal objection runs the other direction here: could there be a structure that is compositionally sterile despite having high I_6 ? This is a genuine possibility, and the paper does not claim I_6 is the only route to generative fertility. The claim is more modest: I_6 is the primary mechanism through which surface-operative constraints generate architectural fertility, and it is the mechanism most amenable to cross-domain measurement. If alternative mechanisms exist, they must be specified with equal precision and subjected to the same empirical tests before the framework can adjudicate between them.

Objection 4: This Is Just Modularity

Modularity theory in evolutionary biology and systems biology addresses the tendency of complex systems to decompose into semi-independent functional units. The surface-constraint principle and I_6 are related to but distinct from modularity. Modularity measures whether a system has separable functional components. I_6 measures whether a structure's boundary presents diverse compositional opportunities. A highly modular system can have low I_6 if its modules combine in a small number of standardized ways. Modularity is a structural property of systems; I_6 is a combinatorial property of individual structures within those systems. The

empirical test is whether I_6 adds predictive power beyond existing modularity indices for architectural fertility. The prediction is that it does, because modularity does not distinguish between systems whose modules combine productively and those whose modules combine but produce limited higher-order structures.

Conclusion

This paper has proposed interface dimensionality as the mechanism underlying the TNDS variable Γ and has argued that this proposal resolves three outstanding problems in the Γ lineage while generating new falsifiable predictions. The core results are as follows.

The circularity problem is resolved by defining I_6 in terms of the constraint rules of the current architecture applied to the structure's present form, without reference to generative history or future participation. The representation problem is addressed by the argument that interface counts are more stable across encoding schemes than raw structure counts, a prediction that is directly testable. The independence problem is resolved by identifying a genuine orthogonality between S^c and I_6 : S^c measures stability well depth while I_6 measures compositional horizon breadth, and the four-quadrant case analysis shows that neither determines the other.

The surface-constraint principle provides a mechanism for why some constraints are architecturally fertile. Constraints that operate at the interface layer of structures rather than at their internal composition layer simultaneously enforce reproductive specificity and preserve contextual abstractness. This combination is what generates high interface dimensionality. The principle is demonstrated across chemistry, language, mathematics, and scientific institutions, and it generates the same qualitative prediction in each domain: the most productive architectures are those whose core constraints are surface-operative.

The paper's central wager is that the question driving the TNDs lineage for several stages, why are some architectures compositionally fertile while others are sterile, can now be stated precisely: what structural property of constraints generates broad compositional horizons in their outputs? The surface-constraint principle proposes an answer. Whether that answer is correct is an empirical question, and the predictions in Section 6 are designed to make it one. The next move belongs to experiment.

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